



BotLab Dynamics' light show drone



Dr Sarita Ahlawat, MD & Co-Founder, BotLab Dynamics

platforms, electronics, and control software for defence and commercial applications.

Reflecting on the company's progress, Sarita explains that by 2018-19, BotLab Dynamics had demonstrated its swarm technology to senior defence leadership. A defining shift followed during field tests in harsh terrains such as Pokhran, where real-world conditions revealed a critical gap. While the software performed well, the reliance on imported, hobby-grade hardware led to repeated failures under extreme heat, dust, and strong winds typical of Indian operating environments.

This realisation prompted a major strategic shift in 2019. "We

decided to rebuild everything from scratch," Sarita explains. "We began designing our own PCBs, selecting and integrating sensors internally, and aligning the hardware very closely with our in-house algorithms." Core components, including flight controllers, motor controllers, and GPS systems, were

redesigned in-house to improve performance, durability, and cost-efficiency, making large-scale swarm operations commercially viable.

Today, BotLab Dynamics' drones are purpose-built for swarm applications. Each unit supports synchronised flight, real-time communication, and precise positioning. A proprietary ground control station continuously communicates with thousands of drones during flight, issuing corrections and maintaining synchronisation. "Every drone knows exactly where it is in three-dimensional space, which allows us to safely execute complex formations at scale," Sarita notes.

For large-scale drone light shows, the company prioritises safety and predictability. "We do not use onboard AI for light shows," Sarita explains. "Instead, we rely on detailed simulations and pre-coded flight paths. Every show is first tested in a virtual environment that accounts for wind and other variables, and we also create digital geofences so drones stay within a defined volume."

Alongside entertainment applications, the startup is advancing defence-focused drone technologies through its Vayudh division. This portfolio includes both small and large drones designed to carry

higher payloads, where onboard intelligence is critical. "For defence platforms, autonomy is essential," Sarita says. "We are developing AI and computer vision systems that allow drones to take decisions on board, without depending on constant ground intervention."

BotLab Dynamics claims to have executed over 300 drone shows across India and internationally, flying 3500 drones in 2023 and 5500 in 2024, placing the company among the top five globally in large-scale swarm operations. It now operates in more than ten countries, including Africa, the Middle East, and Sri Lanka, with plans for even larger swarm demonstrations. In 2025, the company added two more Guinness World Records—one at the Mysuru Dasara Festival and another at the Telangana Rising Summit—bringing its total to seven.

While drone show services currently account for the bulk of its revenue, the startup has begun commercialising its drones, components, and simulators. Its manufacturing ecosystem spans in-house design, SMT collaborations in Manesar and Noida, and assembly and testing operations in Okhla, with certifications managed through authorised agencies. BotLab Dynamics employs approximately 240 people.

Despite its progress, challenges remain. "Finding skilled hardware, RF, and systems engineers continues to be difficult, and scaling manufacturing in India is still a challenge," Sarita says, highlighting the realities of building deep-tech hardware in the country.

—Nitisha Dubey

3 BUILDING COUNTER-DRONE SYSTEMS FOR A NEW ERA OF AERIAL THREATS

An Indian startup fights back with smart technology, protecting sensitive locations from increasingly stealthy drones with advanced detection and neutralisation.

By 2018, drones were changing fast. Once limited to student projects and event photography, drones could now fly